

Neutron Diffraction Residual Strain Measurements

1. Overview

Neutron diffraction residual strain measurements were conducted on AM Bench 2018 test artifacts produced using laser powder-bed fusion additive manufacturing. The measurements were conducted for an IN625 specimen and a 15-5 stainless steel specimen. Figure 1 indicates the measurement positions for lattice strains on the X-Z plane centered on the midplane of the specimen. Strains along the X, Y, Z directions were measured using neutron diffraction at the NIST Center for Neutron Research (NCNR) using the BT8 diffractometer. The measurement gauge volume is approximately a cube with edges that measure 1.5 mm.

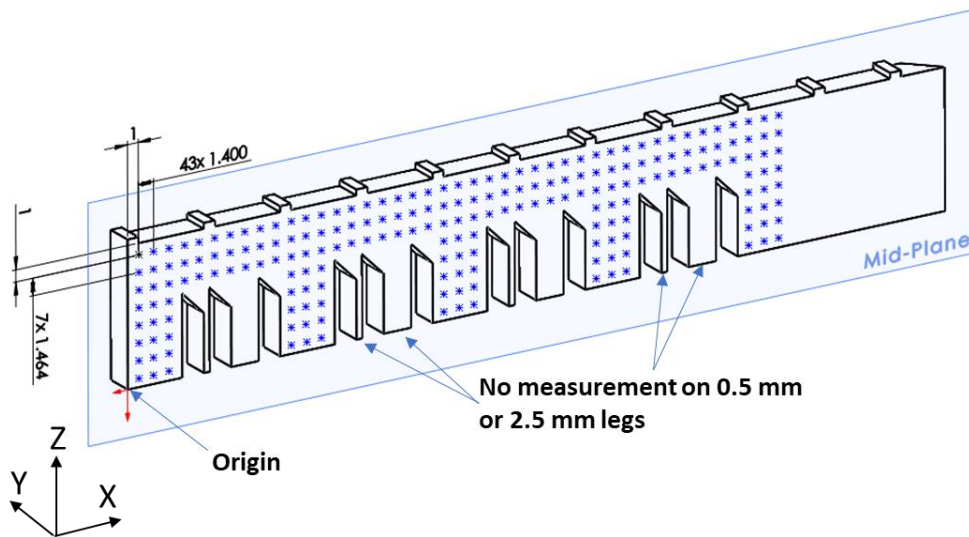


Figure 1. Measurement locations for strain values measured at the NCNR.

2. Summary Results

Figure 2 shows the lattice strain components along the X, Y, and Z directions for the IN625 test artifact. Figure 3 shows the lattice strain components along the X, Y, and Z directions for the 15-5 stainless steel test artifact.

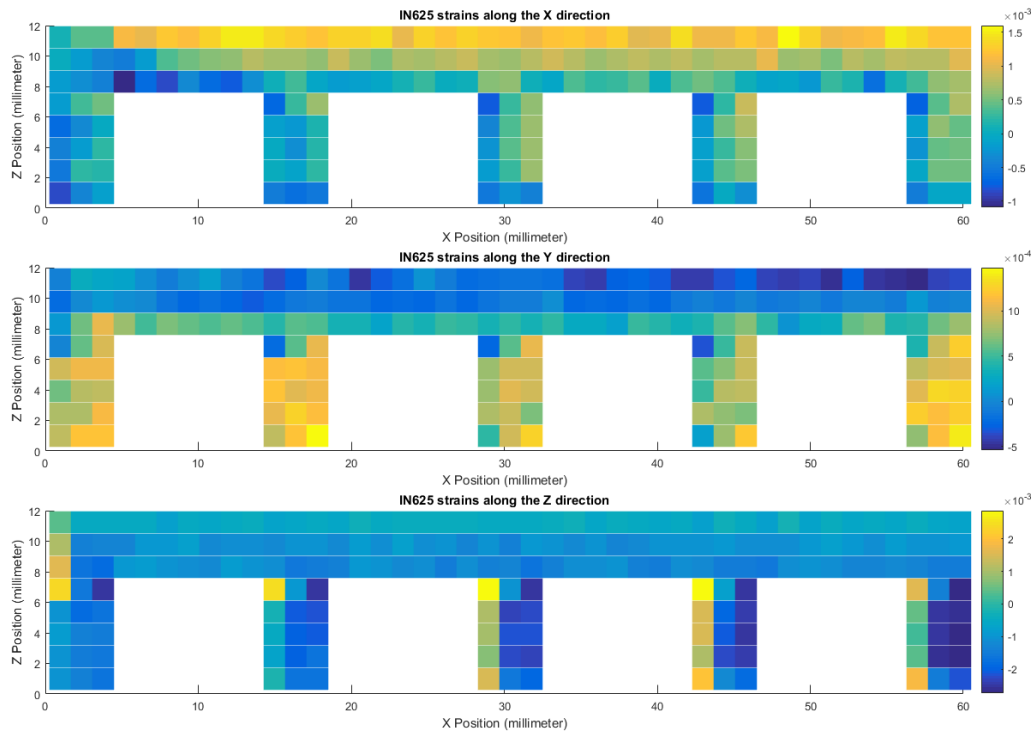


Figure 2. Residual strain maps for the X, Y, and Z directions for the IN625 sample.

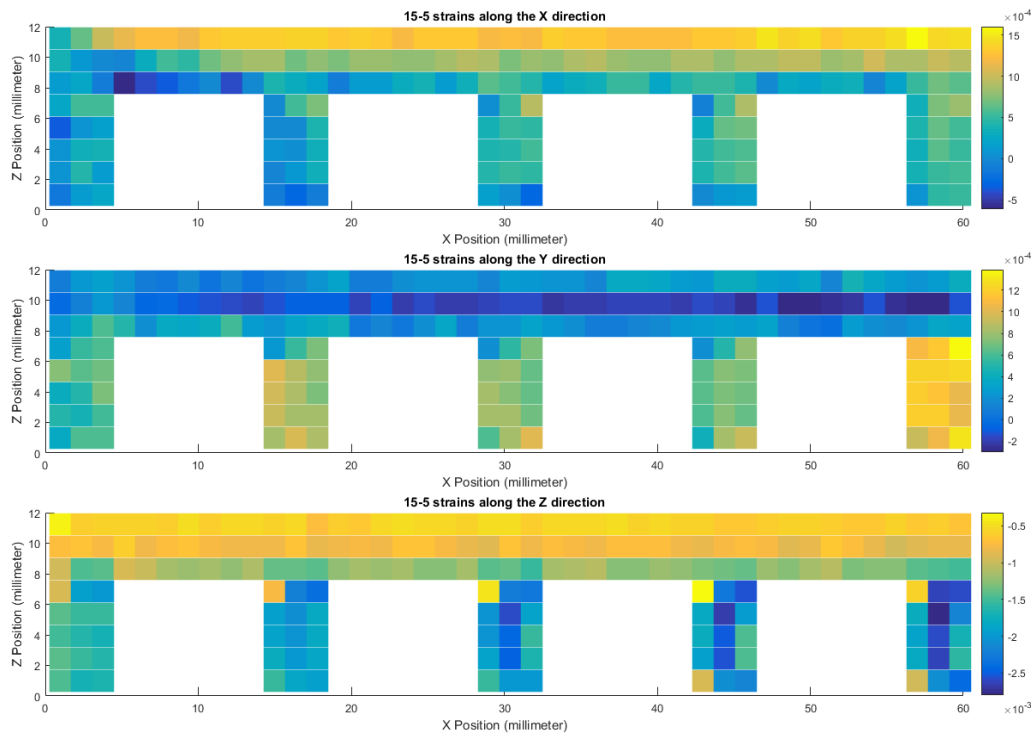


Figure 3. Residual strain maps for the X, Y, and Z directions for the 15-5 stainless steel sample.

3. Data Files

The 15-5 and IN625 neutron diffraction lattice strain measurement results are found in two separate folders. Within each folder, the lattice strain components along the X, Y, and Z directions for each specimen are presented in a data table as a text file. Each text file contains five columns, indicating the X position, Z position, ϵ_{yy} , ϵ_{xx} , and ϵ_{zz} , respectively. The X and Z directions follow the convention established by Figure 1. The X and Z positions indicate the location (in millimeters) of the **center** of the measurement gauge volume, with the origin set at the bottom left corner of the X-Z plane.

Note that the measurements in the legs are not symmetric about the center of the legs in the X direction. Looking at the X-Z plane, with the Y direction into the page, the left most measurement locations in the legs are closer to the left Y-Z surface. Measurement gauge volumes' centers are 1 mm from the Y-Z surface for the left side of the legs, whereas the measurement gauge volumes' centers are 1.2 mm from the Y-Z surface for the right side of the legs.

In addition, each of the results plots in Figure 2 and Figure 3 are provided as high resolution Portable Network Graphics (*.png) image files, and Matlab figure (*.fig) files.